

# Lab 01: linear algebra + computing review

Ikea furniture

Your name + Your teammates' names

2026-08-31

## Setup

Load packages and data:

```
library(tidyverse)
library(tidymodels)
ikea <- read_csv("data/ikea.csv")
```

## Linear algebra review

knitr::include\_graphics("exercise1.jpg")

1. a:  $3 \times 2$       2. a) is ii, This is the only combination where row <sup>dimensions</sup> match with column.

b:  $2 \times 4$

c:  $2 \times 3$       b)  $\begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 2 & 2 & 2 \end{bmatrix} \times \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$

d:  $m \times p$

e:  $n \times n$        $\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} \times \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 3 & 5 & 7 \\ 3 & 7 & 11 & 15 \\ 5 & 11 & 17 & 25 \end{bmatrix}$

3. a: True, same dimensions, add all corresponding elements together.

b: True, This is the distributive law of multiplication.

c: False. This is a weird form of distributive law. ~~Matrix~~ Matrix multiplication is irreversible.

4. a)  $X$  is non-singular. The dimension is not  $n \times n$  but  $n \times p$ .

b) We don't have enough condition to determine if  $X^T X$  is singular. This is because ~~if~~ we don't know if the row or column is depend linearly dependent or not.

5. a)  $a-b = \begin{bmatrix} a_1 - b_1 \\ \vdots \\ a_n - b_n \end{bmatrix}$       b. a)  $X_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$   $X_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$   $X_3 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$(a-b)^T(a-b) = \sum_{i=1}^n (a_i - b_i)^2$       We want:  
 $a_1 X_1 + a_2 X_2 + a_3 X_3 = 0$  however we can't find corresponding  $a_1, a_2, a_3$   
 $\therefore$  This is a full column rank matrix.

b)  $X_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ ,  $X_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ ,  $X_3 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

We want:  
 $a_1 X_1 + a_2 X_2 + a_3 X_3 = 0$ . We can't find corresponding  $a_1, a_2, a_3$ .  
 $\therefore$  This is a full column rank matrix.

c)  $X_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ ,  $X_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ ,  $X_3 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ ,  $X_4 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

We want:  
 $a_1 X_1 + a_2 X_2 + a_3 X_3 + a_4 X_4 = 0$ .  
 let  $a_1 = 1, a_2 = -1, a_3 = -1, a_4 = 1$ :  
 $\begin{bmatrix} 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 0$   
 $\therefore$  This is not a full column rank matrix.

The answer 1e changed into  $P^*P$ .

## Computing review

### Exercise 7

```
ikea <- read_csv("data/ikea.csv")
```

Rows: 3694 Columns: 13

-- Column specification -----

Delimiter: ","

chr (6): name, category, link, other\_colors, short\_description, designer

dbl (6): X1, item\_id, depth, height, width, price\_usd

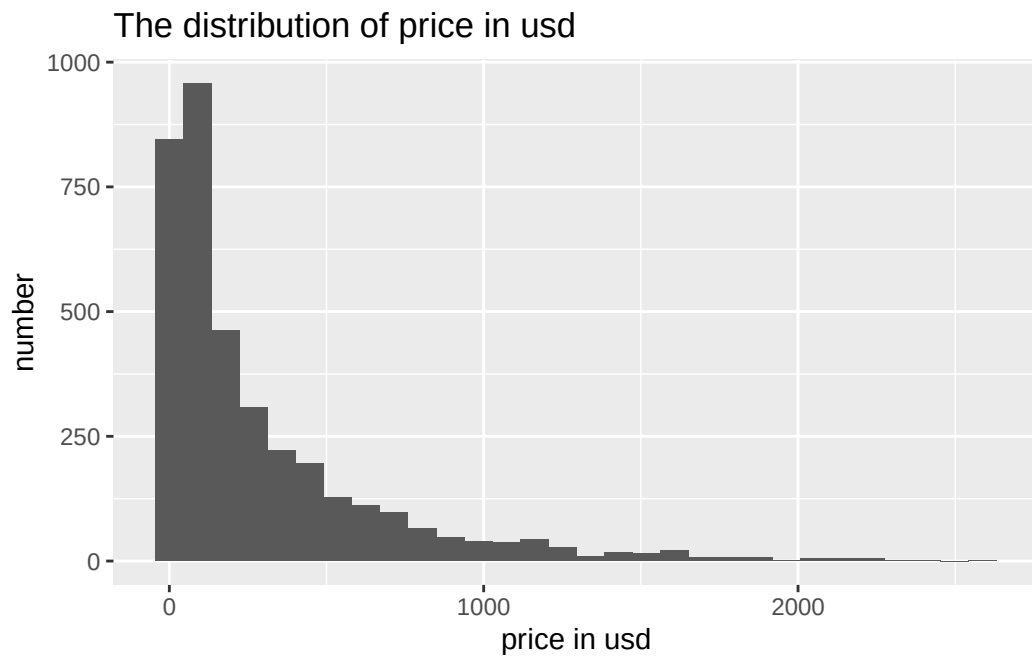
lgl (1): sellable\_online

i Use ``spec()`` to retrieve the full column specification for this data.

i Specify the column types or set ``show_col_types = FALSE`` to quiet this message.

```
ggplot(data = ikea, aes(x = price_usd)) +  
  geom_histogram() +  
  labs(x = "price in usd",  
       y = "number",  
       title = "The distribution of price in usd")
```

``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



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**Exercise 8**

The distribution is skewed to the left, and there are some outliers that will affect the mean to be larger than the median.

## Exercise 9

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## Exercise 10

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